MMAN4410 Finite Element Methods

Brake Disc Final Report

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Executive summary

This project evaluates the thermal, structural, and vibrational performance of a ventilated stainless-steel motorcycle brake disc using Finite Element Method (FEM) analysis. The model is based on a Wilwood-style rotor, with loading conditions representing a 250 kg motorcycle and rider descending a 1 km, 10° grade under continuous braking.

Energy-based calculations yielded a braking power of 5.9 kW, with 2.66 kW absorbed per rotor, and a corresponding frictional force of 251 N. Steady-state thermal, static structural, modal, and harmonic analyses were performed in ANSYS using a 1/10th cyclic-symmetry sector to reduce computational cost.

Both the full-disc and sector models predicted a peak temperature of approximately 257.5 °C at the outer friction track, consistent with published values for severe braking and within the operating limits of AISI 420 stainless steel and sintered pads. Structural stresses were dominated by thermal expansion, with a converged average stress of about 96 MPa and a peak of 347 MPa occurring at cooling-hole edges, attributable to geometric stress concentrations.

Modal and harmonic analyses showed no structural resonances within the 1–16 kHz brake-squeal band, indicating low risk of rotor-induced squeal for the given load case.

Overall, the disc demonstrates acceptable thermal and vibrational performance, with the main limitation being elevated local stresses around cooling holes. Future improvements should focus on refining hole geometry, boundary conditions, and transient thermo-mechanical effects.

1 Introduction and Aims

Automotive braking systems are critical components that directly affect vehicle safety and performance. However, disc brakes are subject to high thermal and mechanical stresses during operation, which can lead to performance degradation, brake fade, or even structural failure. Furthermore, vibrational instabilities known as brake squeal not only reduce user comfort but can also indicate undesirable dynamic coupling between the rotor and pads.

The main objective of this project is to investigate and improve the performance of a ventilated disc brake using finite element analysis (FEA). The disc brake will be considered for a motorcycle going down a 1km mountain at a 10 degree grade. The study focuses on two primary aspects:

Thermal Performance: Ensuring efficient heat dissipation during braking cycles to minimise thermal stresses, prevent material degradation, and maintain consistent braking performance.

Vibrational Behaviour: Analysing modal characteristics of the rotor to identify and mitigate potential brake squeal frequencies that can compromise noise, vibration, and harshness (NVH) performance.

To achieve these aims, a ventilated stainless steel disc rotor with typical motorbike dimensions will be modelled. Both thermal, structural and vibrational analyses will be conducted using ANSYS Workbench, incorporating realistic operational and boundary conditions such as force from the brake pads, frictional heating, and convective cooling.

This investigation will provide insight into the coupled thermo-mechanical behaviour of disc brakes and guide design optimisations to enhance durability, efficiency, and user experience in automotive braking systems.

Why Finite Element Analysis is Used

Finite Element Analysis (FEA) is used in this project because it provides a practical way to predict the stress and temperature fields in the brake disc under realistic loading, without relying solely on physical testing. As summarised by NAFEMS and the Open University [16], FEA offers several key advantages:

- improved product performance and cost-effectiveness through virtual design changes rather than trial-and-error hardware,
- reduction of development time and the amount of experimental testing required,
- first-time achievement of required quality and safety margins,
- better information for engineering decision-making and satisfaction of design codes and contractual requirements.

The same source emphasises that all FEA models are approximate and that their accuracy depends on understanding the behaviour of the system being modelled and on making appropriate assumptions for loads, restraints, and material properties [16]. For a brake disc, determining stresses and strains at all points in the rotor, in all directions, for all combinations of thermal and mechanical loading is not feasible with hand calculations. FEA is therefore used to model the disc under the specified downhill braking scenario, with carefully defined load cases, boundary conditions, and temperature-dependent properties, so that the resulting stress, temperature, and modal predictions can be used to assess performance against the quantitative goals in Table 1.

Quantitative Project Goals

The following measurable goals have been established to evaluate the rotor's performance and verify model validity:

Table 1: Quantitative performance goals for the ventilated brake disc analysis.

Category	Metric / Goal	How it'll be measured	Rationale / Source
Thermal limit	$T_{\max} \leq 350\,°\mathrm{C}$ at the friction track during peak braking	Extracted from steady-state thermal simulations (max temperature probes)	OEM motorcycle brake pads begin to fail/fade at this temperature [8] (Not the Stainless Steel).
Structural stress	$\sigma_{\max} \leq 345$ MPa	Obtained from static structural analysis with imported thermal fields	Tensile Yield Strength of AISI 420 Stainless [2].
NVH / squeal risk	No rotor natural frequency within $\pm 10\%$ of pad-pass frequency in the 1–16 kHz squeal band	Modal analyses and comparison with calculated pad-pass frequency	SAE J2521 [9] defines squeal events in this frequency range; detuning minimises excitation risk.
Convergence (mesh)	$\Delta T_{\max} \leq 5\%$, $\sigma_{\max} \leq 10\%$ between successive mesh refinements	Six mesh densities; results plotted versus element count	Standard verification criterion to confirm mesh independence.

2 Assumptions and Problem Specifics

For the purposes of this report, the load case will be considered as a motorcycle and rider descending a total elevation of 1000 m down a long mountain road. The average road incline is taken as 10°, and the motorcycle maintains a constant speed of 50 km/h while continuously applying the brakes to control the descent.

2.1 Assumptions

Some key assumptions for estimating the inputs and the boundary conditions are as follows:

- Aerodynamic drag, rolling resistance, and engine braking are neglected (will be significant in reality, done for simplicity).
- The gravitational potential energy loss is entirely absorbed by the braking system.
- Combined mass of motorcycle and rider: $m = 250$ kg.
- Constant velocity: $v = 50$ km/h = 13.89 m/s.
- Average road grade: $\theta = 10°$.
- Brake heat is evenly distributed between the front and rear wheels (50:50 split).
- Each axle has a single disc brake (one front and one rear).
- Ninety percent of the braking energy is absorbed by the rotors ($\eta = 0.9$).
- A wheel radius of 0.3m
- Ambient temperature of $T_\infty = 25$ °C

2.1.1 Heat Flow

The aim of this section is to estimate the **heat flow (power)** that must be dissipated by the brakes and ultimately conducted into the brake discs during this steady downhill braking condition.

1. Road length and descent time

$$L = \frac{h}{\sin\theta} = \frac{1000}{\sin 10°} = 5759 \text{ m}$$

$$t = \frac{L}{v} = \frac{5759}{13.89} = 415 \text{ s} \approx 6.9 \text{ minutes}$$

2. Power required to maintain constant speed

$$P_{\text{brake}} = mgv\sin\theta$$

$$P_{\text{brake}} = 250 \times 9.81 \times 13.89 \times \sin 10^\circ = 5.9 \text{ kW}$$

3. Distribution of heat flow Each wheel carries half of the total braking power:

$$P_{\text{wheel}} = 0.5 \times P_{\text{brake}} = 2.95 \text{ kW}$$

The fraction of energy entering each rotor is then:

$$P_{\text{rotor}} = \eta\, P_{\text{wheel}} = 0.9 \times 2.95 = 2.66 \text{ kW}$$

Each rotor has two friction faces, so the heat flow per face is:

$$P_{\text{face}} = \frac{P_{\text{rotor}}}{2} = \frac{2.66}{2} = 1.33 \text{ kW}$$

2.1.2 Tangential Braking Force

The braking torque generated by a disc brake can be expressed as

$$T = 2\,\mu\,F_N\,R_{\text{eff}},$$

where T is the brake torque, μ is the pad–disc friction coefficient, F_N is the total normal clamping force applied by both pads, and R_{eff} is the effective friction radius corresponding to the centroid of the pad contact band. For the present rotor geometry, $R_{\text{eff}} = 0.127$ m.

The braking torque per wheel is obtained from the required braking power. Using $P_{\text{brake}} = 5.9$ kW, a wheel radius of $R_{\text{wheel}} = 0.30$ m, and a descent speed of $v = 13.89$ m/s, the wheel angular velocity is $\omega = v/R_{\text{wheel}}$. The torque required at each wheel is therefore

$$T_{\text{wheel}} = \frac{P_{\text{brake}}}{2\,\omega} = \frac{P_{\text{brake}}}{2(v/R_{\text{wheel}})} = 63.7 \text{ N·m}.$$

Taking a friction coefficient of $\mu = 0.40$ [5], the corresponding total normal clamping force on the disc is

$$F_N = \frac{T_{\text{wheel}}}{2\,\mu\,R_{\text{eff}}} = \frac{63.7}{2 \times 0.40 \times 0.127} \approx 627 \text{ N}.$$

The frictional (tangential) braking force acting on the disc is then

$$F_{\text{fric}} = \mu\,F_N = 0.40 \times 627 \approx 251 \text{ N}.$$

2.1.3 Convection

While descending the mountain at constant speed, the brake disc is fully exposed to ambient airflow, and convective heat transfer plays a significant role in removing heat from its surfaces. Both faces of the rotor, as well as the vane passages, exchange heat with the surrounding air through forced convection as the wheel rotates. Given the open geometry and high airflow around the motorcycle brake, convection can be approximated using an empirical film coefficient h applied to all external surfaces.

A research article with experimental measurements on ventilated steel disc brakes reported average convective heat transfer coefficients of 27.0, 52.7, and 78.3 W/m^{2}K for vehicle speeds of 20, 40, and 60 km/h respectively [6]. As the present analysis represents a descent at approximately 50 km/h, a representative value of about $h_1 = 60$ W/m^{2}K was adopted for the inner hub surface of the rotor and $h_2 = 80$ W/m^{2}K was adopted for the section of the rotor with the cooling holes in steady-state thermal simulations.

2.1.4 Radiation

In addition to convection, thermal radiation contributes to heat loss from the rotor, particularly when surface temperatures are elevated. The exposed surfaces of the brake disc radiate heat to the surrounding environment. For oxidised stainless steel surfaces similar to the AISI 420 material used in this study, emissivity values in the range of 0.7–0.8 have been reported [7]. A nominal emissivity of $\varepsilon = 0.75$ was therefore adopted, representative of a lightly oxidised stainless-steel rotor surface.

2.2 Summary of Initial Conditions

To summarise, the applied loading and boundary condition parameters used in the thermal and structural analyses are compiled in Table 2. These values represent the steady downhill braking condition for a single wheel, based on a constant descent speed of 50 km/h and a 10° road grade. The parameters encompass the thermal input from frictional heating, convective and radiative cooling, and the mechanical pressure generated by pad contact.

Table 2: Summary of applied thermal and mechanical parameters for one wheel.

Parameter (per rotor)	**Value / Unit**
Heat Flow	$P_{\text{rotor}} = 2.66$ kW
Convection coefficient	$h_1 = 60$ W/m^{2}K $h_2 = 80$ W/m^{2}K
Radiation emissivity	$\varepsilon = 0.75$
Normal brake force	$F_N = 627$ N
Tangential brake force	$F_T = 251$ N

3 Model Definition

3.1 Models and Governing Equations

The finite element workflow (Fig. 1) consists of linked steady-state thermal, static structural, modal, harmonic response, and harmonic acoustics analyses.

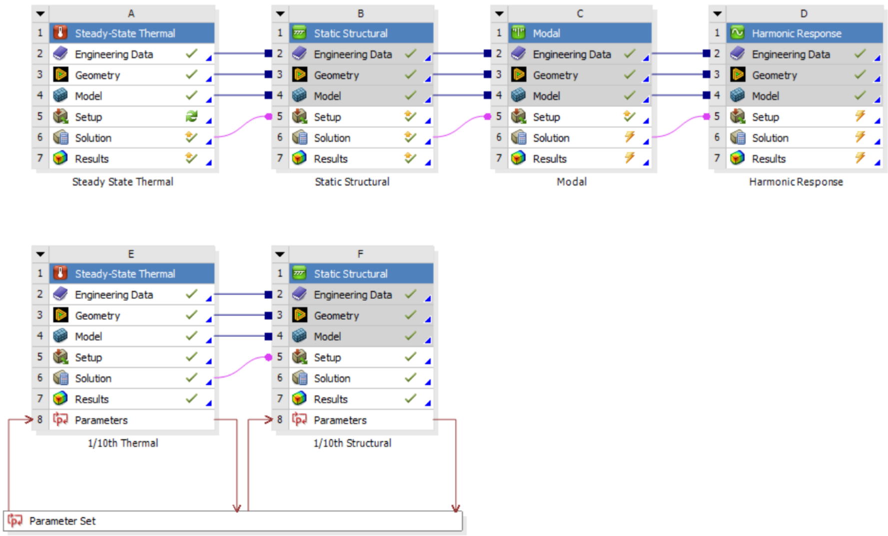


Figure 1: Workbench Models

These models are governed by standard conservation laws for heat transfer, solid mechanics, and vibration.

Thermal model: Steady-state heat conduction is solved using Fourier's law,

$$\nabla \cdot (k\nabla T) + q = 0,$$

with prescribed heat flux on the friction surfaces to represent braking, and convection–radiation boundary conditions on exposed areas. The resulting temperature field defines the thermal load for subsequent analyses.

Structural model: The static structural analysis applies thermoelastic equilibrium,

$$\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{F} = 0, \quad \boldsymbol{\varepsilon}_{\text{thermal}} = \alpha(T - T_0)\mathbf{I},$$

to determine induced stresses and deformations under mechanical and temperature loads.

Vibration and acoustics: Modal and harmonic response analyses solve the equation of motion,

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{F}(t),$$

to extract natural frequencies and steady-state responses under pad excitation. The harmonic acoustics model then predicts radiated sound pressure using the Helmholtz equation.

Computational efficiency: To reduce model size, shell elements are used to model the full braking case while a 1/10th sector of the disc is simulated using cyclic symmetry boundary conditions to analyse mesh convergence and support the primary model. These choices ensure the solution accurately represents full-disc behaviour while minimising element count and runtime.

3.2 Brake Rotor Geometry

The geometry of the brake disc was chosen with the aim of meeting the project goals defined in Table 1. The brake geometry for this report is loosely based on a motorcycle rotor from Wilwood Disc Brakes [3] Appendix A.1, a proven design in real life. In this study, the following key geometric parameters, seen in the table below, were used to define the brake disc model for analysis.

Table 3: Geometric parameters of the motorcycle brake disc.

Parameter	Symbol / Description	Value (mm)
Number of bolt holes	N_v	10
Outer diameter	D_o	290
Inner diameter	D_i	50
Disc thickness	t_p	6
Hole Sizes	H_h	9

These dimensions were selected to maintain geometric and thermal similarity to a realistic ventilated rotor while omitting complexities to remain computationally efficient for simulation.

Based on the geometric parameters and following the motorcycle rotor design, the final geometry of the brake disc was modelled below using onshape (Fig 3).

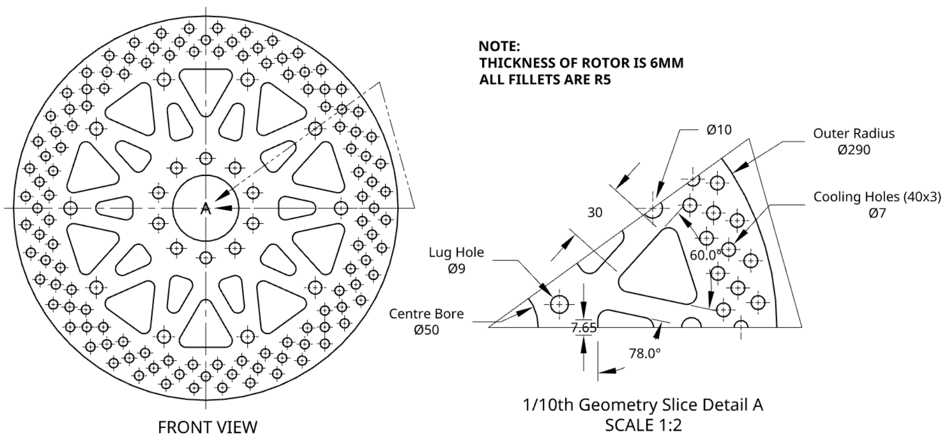


Figure 2: Brake Disc Drawing

The full brake assembly is defined in a cyclical coordinate system. It was modelled using onshape, and has slight modifications to its real world counterpart, such as the addition of more lug holes to allow for the ability to use cyclic symmetry.

3.3 Brake Material Properties

AISI 420 stainless steel was selected as the brake disc material due to its common use in motorcycle brakes [4]. This steel provides a balance of high strength, wear resistance, and moderate thermal conductivity, making it well-suited for the heating and cooling experienced during braking. Its structure offers good hardness and friction characteristics when paired with sintered or organic brake pads, while maintaining corrosion resistance superior to grey cast iron.

Table 4: Material properties for AISI 420 martensitic stainless steel [1, 2]

Property	**Value**
Density, ρ	7700 kg/m^3
Young's modulus, E	200 GPa
Poisson's ratio, ν	0.27
Ultimate tensile strength (UTS)	655 MPa
Yield strength (0.2% proof)	345 MPa
Brinell hardness, HB	$\leq$444 HB
Thermal conductivity, k	24.9 W/m·K
Specific heat capacity, c_p	460 J/kg·K
Coefficient of thermal expansion, α	10.8 μm/m·K

3.4 Model Setup

The FEM model considers a full disc including the frictional force from the brake pads and the heat flow due to that. A 1/10th disc is used to support this model, however, it only models the effects of the input heat flow and is used for validating mesh convergence.

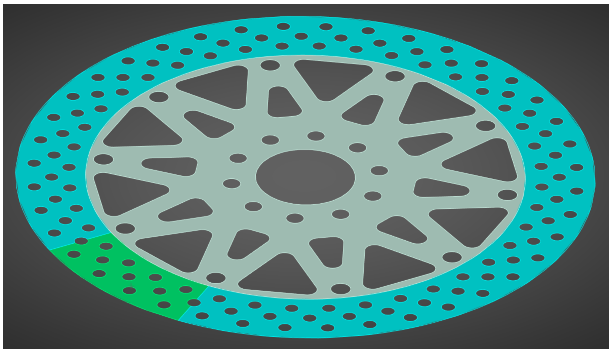


Figure 3: Brake Disc Setup

The area highlighted in green corresponds to the pad–disc contact band where the braking force is applied. Regions highlighted in blue and green receive heat flux. The same face selections are applied on the underside of the disc.

Table 5: Thermal Load and boundary conditions used for the brake disc simulations.

Region / Surface	Type & Value	Application / Notes
Pad contact band	Frictional Load = 251N	Applied over the brake contact band shown in green.
Heat input regions	Heat flux = 1330W	Applied on friction tracks shown in blue/green.
Blue and Green Bands	Convection, $h = 60\ \mathrm{W\,m^{-2}\,K}$	Applied to those exposed surfaces.
Turquoise Hub	Convection, $h = 80\ \mathrm{W\,m^{-2}\,K}$	Slightly higher convection value due to cooling holes
All faces	Thermal radiation, emissivity $\varepsilon = 0.85$	Radiation applied to outer visible faces.

3.5 Mesh strategy

In the preliminary report, a 3D Solid Element meshing strategy was used; however, since this is a flat plate model, the shell element approach was adopted for computational efficiency.

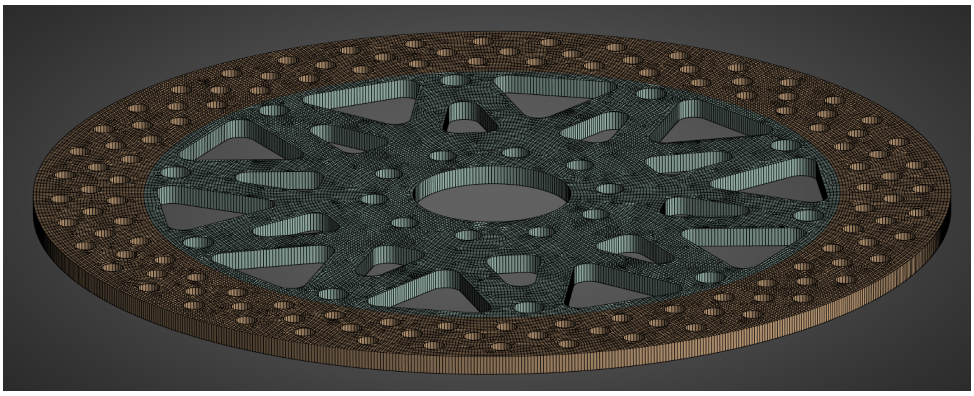


Figure 4: Meshing approach for the brake disc model.

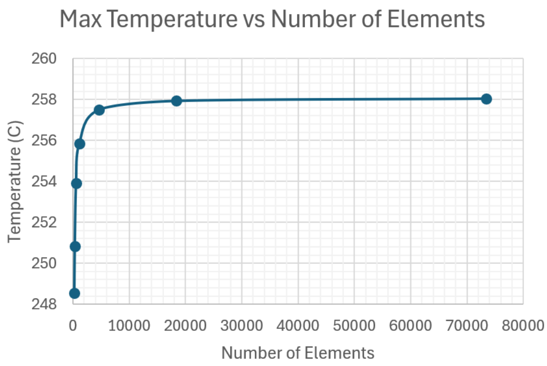
Table 6: Summary of mesh setup and rationale.

Mesh Control	Setting / Value	Purpose / Rationale
Element Type	Shell Elements	Geometry is a simple flat plate, therefore shell elements can be used to reduce computational cost
Global Mesh	Element Size = 1mm	From convergence testing 1mm was adequate for stress and temperature concentrations.

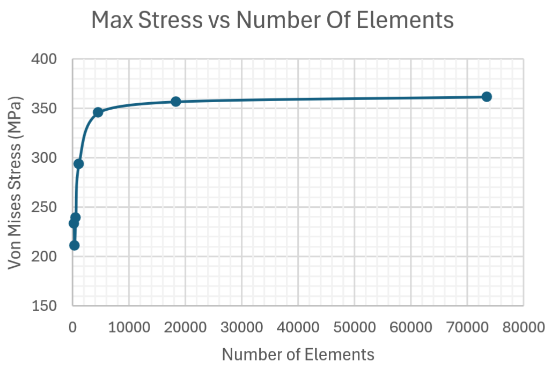
4 Results

4.1 Mesh Convergence

A mesh convergence study was performed using the 1/10th model to evaluate the sensitivity of temperature and stress results to element size. The global element size was varied from 5 mm to 0.25 mm and the full results table from the parameter set can be found in Appendix A.2.



(a) Max temperature vs No. of Elements.



(b) Max Stress vs No. of Elements.

Figure 5: Mesh convergence plots.

Figure 5 shows that both max temperature and max stress begin to converge at around 4000 elements, which is 1mm in global element size.

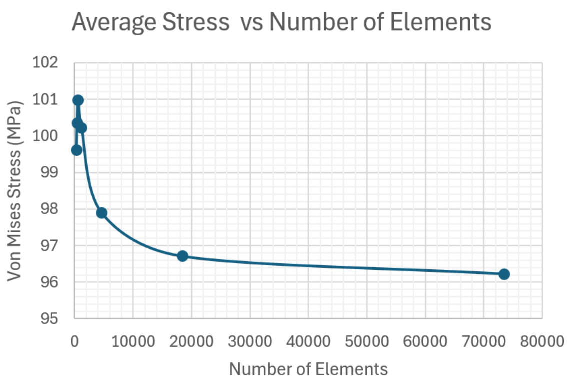
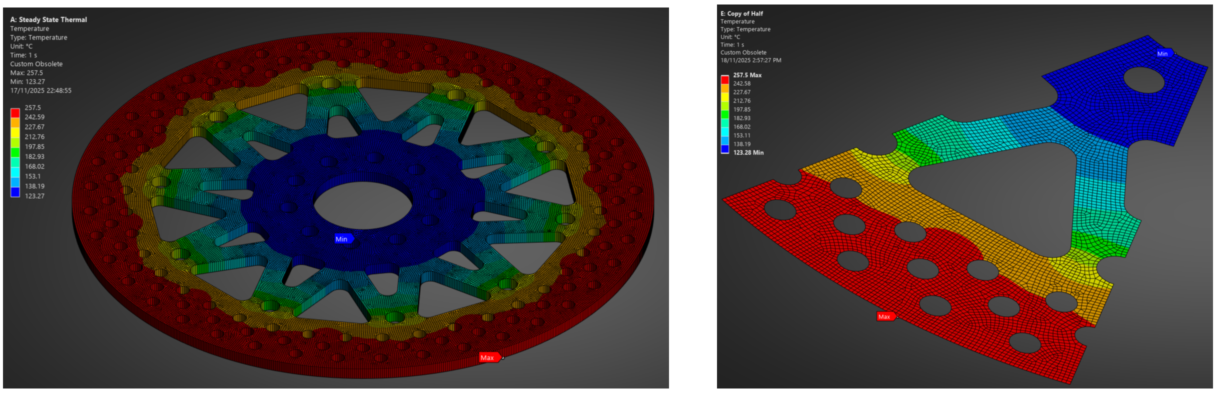


Figure 6: Average Stress vs No. of Elements.

Furthermore, the average stress values (Fig. 6) converged smoothly towards approximately 96 MPa, which remains well below the minimum yield strength of AISI 420 (345 MPa). Therefore, the mesh is considered sufficiently converged, and all subsequent analyses use the 1 mm global element size, as further refinement would increase computational cost without providing significant accuracy improvements.

4.2 Static Thermal



(a) Full-disc thermal model. (b) 1/10th Secondary thermal model.

Figure 7: Steady-state temperature distributions for the full and supporting model.

Both the full-disc model and the secondary results from the reduced 1/10th sector produced identical steady-state peak temperatures of approximately 257.5°C at the outer friction track. This agreement confirms that the cyclic symmetry boundary conditions applied to the sector model correctly reproduce the thermal field of the full rotor. The temperature distribution shows the expected trend for a ventilated motorcycle disc: the highest temperatures occur at the pad contact band, with temperatures decreasing progressively toward the inner vents and hub due to increasing cooling area and reduced heat input.

4.3 Static Structural

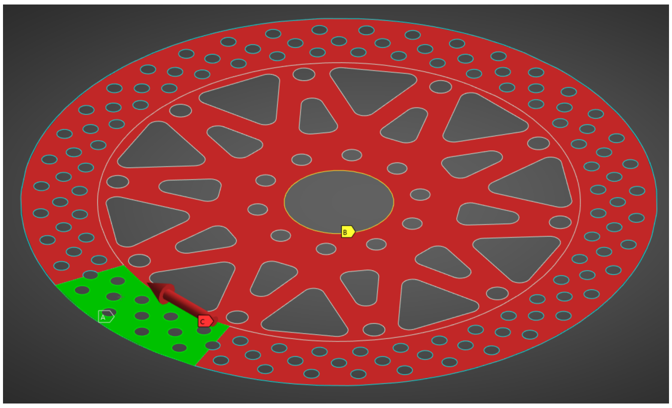
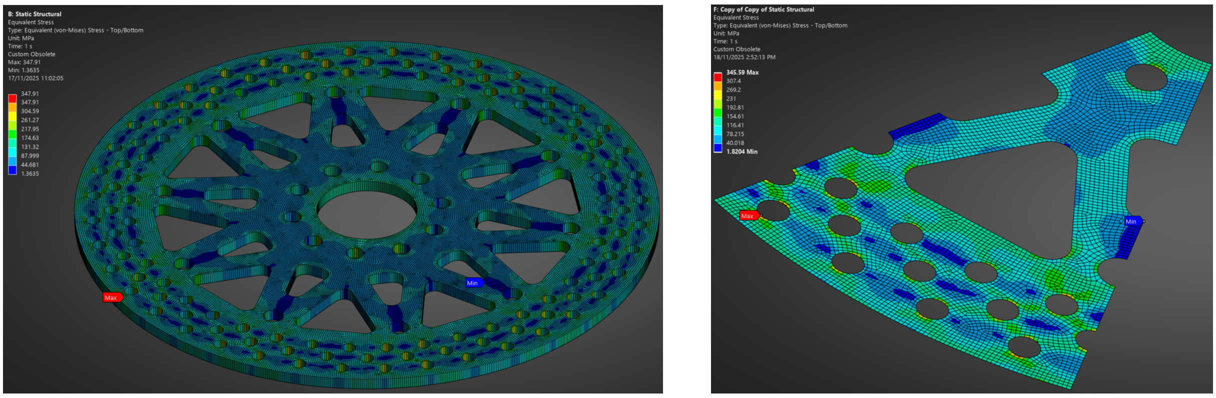


Figure 8: Static structural setup showing brake pad load and displacement conditions.

The static structural analysis applied a tangential frictional force of 251 N on the brake pad contact band to represent the braking load. A pressure load was not used, as the pad force acts symmetrically on both disc faces in practice, and preliminary testing showed that the normal pad pressure had a negligible contribution to the stress field. The steady-state temperature distribution from the thermal analysis was imported to capture thermal expansion and the resulting thermo-mechanical stresses.

To restrain the disc while avoiding unrealistic stress concentrations, the inner bore was constrained using a radial displacement boundary condition. This restricted motion in the global y- and z-directions while still allowing the disc to expand radially, reflecting the physical behaviour of a bolted rotor in which the mounting holes expand slightly under heat. Directly constraining the bolt holes produced severe stress singularities and was therefore avoided.

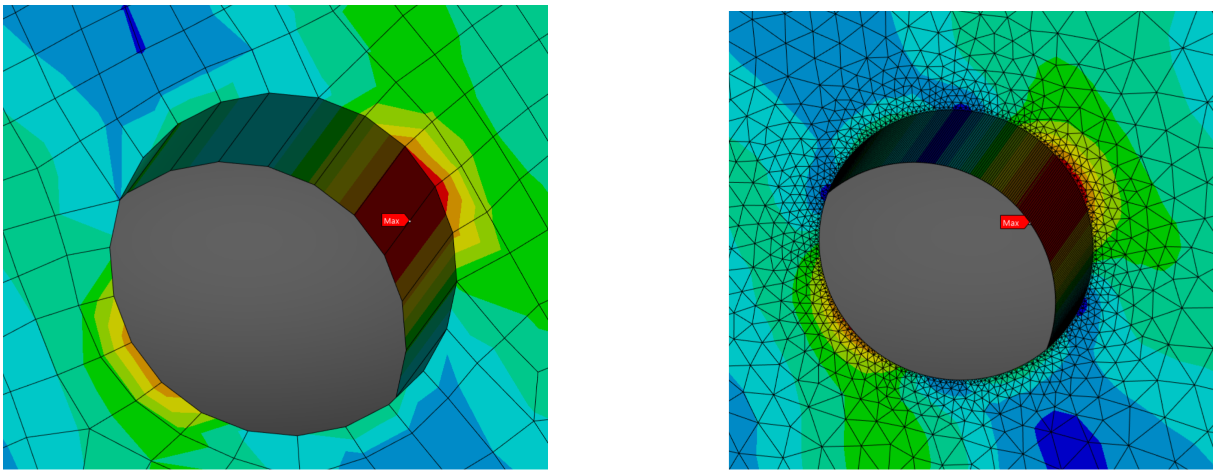


(a) Von Mises stress for full-disc model.

(b) 1/10th Secondary Stress Result.

Figure 9: Primary and Secondary Supporting Von Mises Stress Results.

Although the full-disc model included the frictional shear load from the brake pads and the 1/10th sector model did not, both simulations produced nearly identical stress distributions and magnitudes. This confirms that, for the present load case, thermal expansion dominates the structural response, while the pad force contributes relatively little to the peak stress.



(a) 1 mm global mesh: peak stress hotspot at hole edge.

(b) Locally refined mesh (200 divisions): same hotspot persists.

Figure 10: Comparison of stress distribution near a cooling-hole edge.

The maximum von Mises stress in both models was approximately 347 MPa, occurring at the sharp edges of the outer cooling holes. To verify whether this peak was a genuine geometric concentration or a mesh artefact, the region was re-meshed using an extremely fine local division (200 edge subdivisions). As shown in Fig. 10, both the baseline 1 mm global mesh and the refined mesh produced nearly identical stress peaks at the same location. This confirms that the high value represents a true geometric stress concentration inherent to the hole geometry rather than a numerical error.

Outside this localised singularity, the stress field remains well below the material's yield strength, with converged average stresses of approximately 96 MPa. The stress behaviour is therefore consistent with the expected thermoelastic response of a ventilated disc brake, although the cooling-hole geometry may benefit from refinement to reduce these concentrated peaks.

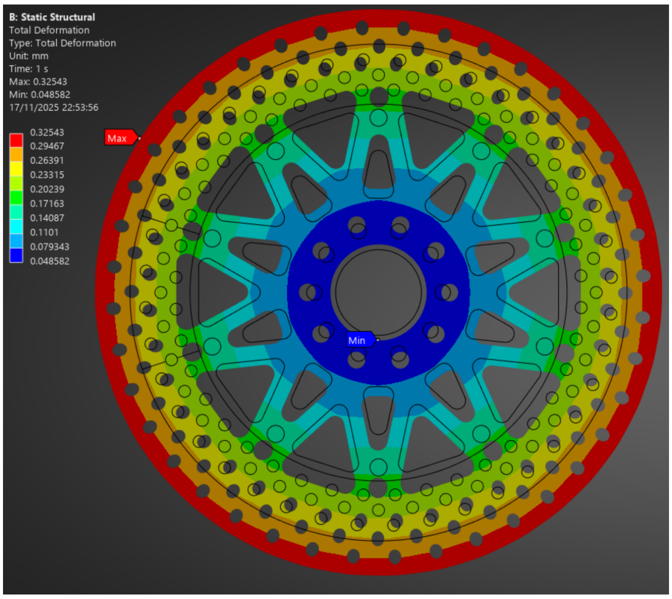


Figure 11: Rotor Deformation

Maximum deformation reached approximately 0.325 mm at the outer edge of the disc, with the overall deflection pattern following the expected radial bending behaviour caused by pad loading and thermal expansion. In addition to this radial expansion, the disc exhibits a slight circumferential deformation in the clockwise direction, consistent with the applied tangential frictional force from the brake pads. Furthermore, around the bolt holes the deformation is only 0.05 mm, which is well within the clearance required for bolt expansion.

4.4 Modal

A modal analysis was performed to identify the natural frequencies and vibration modes of the disc within the brake–squeal range of interest (1–16 kHz). To ensure adequate frequency coverage, 100 mode shapes were extracted, which provided modal solutions up to 16 kHz. This range fully encompasses the frequency band in which brake squeal typically occurs, as defined in SAE J2521 [9].

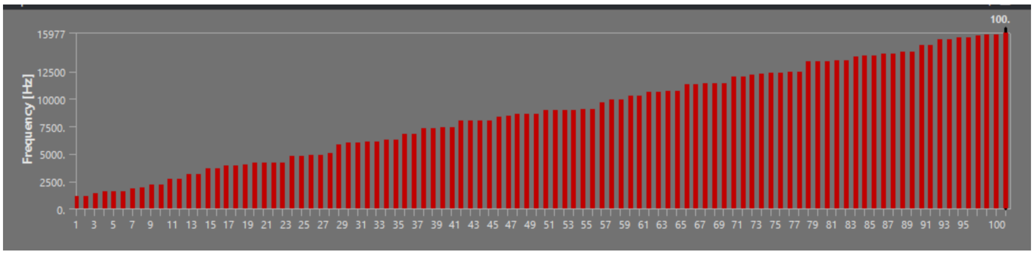
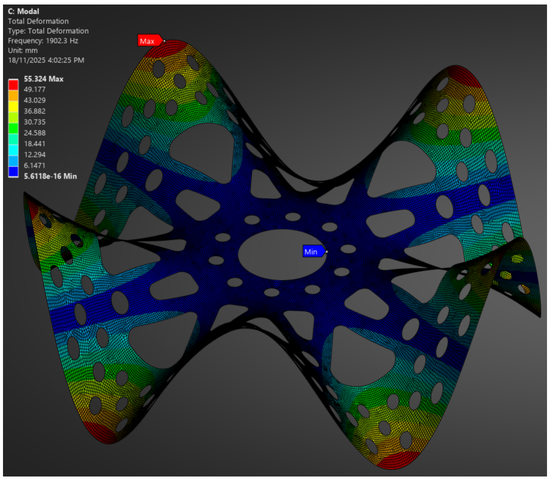


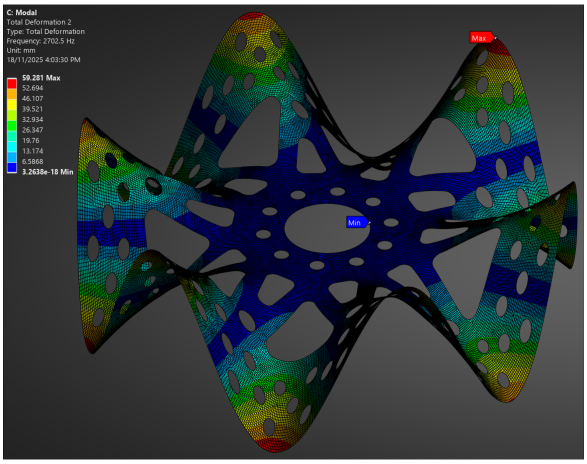
Figure 12: Mode Shapes Betwen 1 to 16 kHz region

From the complete set of modes, a subset of representative mode shapes was selected for presentation. These correspond to frequencies where the harmonic analysis (shown in the next section) showed noticeable amplitude features. Specifically, modes near 1.9 kHz, 2.7 kHz, 8.6 kHz, and 9.9 kHz were chosen because they appeared close to local peaks or troughs in the harmonic response plots.

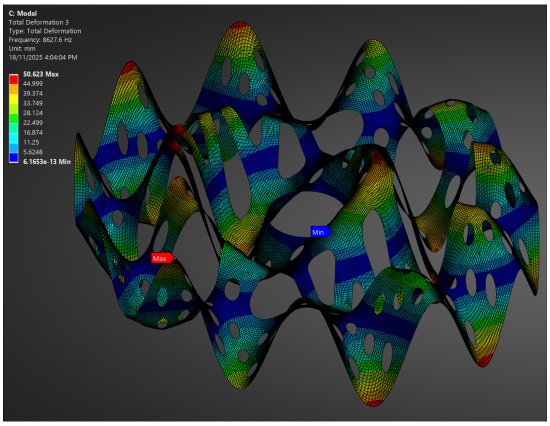
The mode shapes shown in Figures 13(a)–(d) illustrate the progressively increasing complexity of the disc's deformation pattern as frequency increases. The 1.9 kHz and 2.7 kHz modes exhibit relatively smooth, low-order out-of-plane bending with four prominent lobes, producing a wave pattern around the circumference. In contrast, the higher-frequency modes at 8.6 kHz and 9.9 kHz display many more undulations, resulting in a finer deformation pattern.



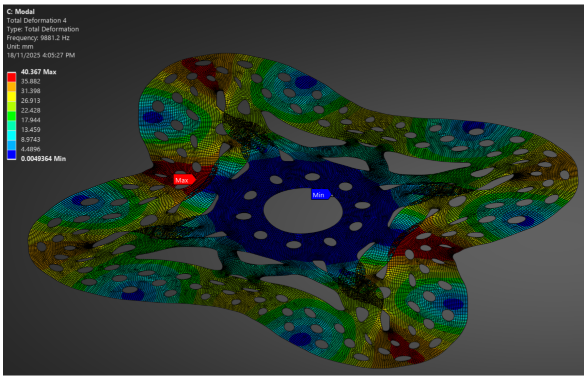
(a) Mode shape at 1.9 kHz.



(b) Mode shape at 2.7 kHz.



(c) Mode shape at 8.6 kHz.



(d) Mode shape at 9.9 kHz.

Figure 13: Significant mode shapes from the modal analysis.

4.5 Harmonic Response

The harmonic response analysis used the modal results obtained from the modal study and applied a small harmonic (0.1 MPa) pressure over the pad–disc contact region. This excitation was monitored from the required full squeal band - 1 kHz to 16 kHz.

Two amplitude clusters were observed in the full-spectrum response (Fig. 14): a broad feature between 1.5–3 kHz and a second feature between 8–10 kHz. These correspond to the frequencies of the mode shapes highlighted in the previous section.

To determine whether these features represented actual resonances, the response in each region was examined more closely (Figures 15 and 16). In both cases, the displacement amplitude varied smoothly with frequency and did not show the sharp, narrow peaks characteristic of structural resonance.

As a result, no significant resonant amplification was observed in the disc across the 1–16 kHz range. Therefore, the likelihood of rotor-driven brake squeal is low for this operating scenario.

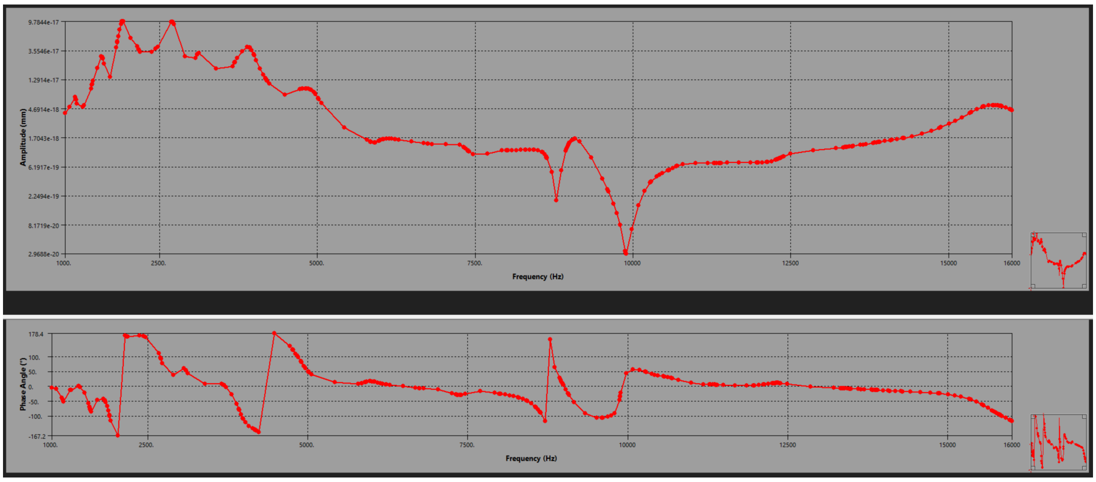


Figure 14: Harmonic Response of Disc from 1 to 16Khz

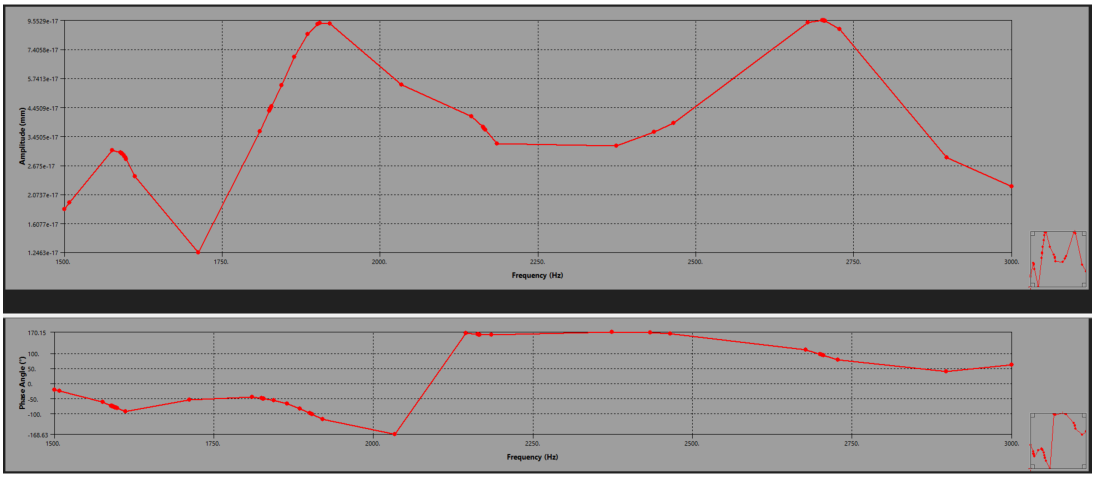


Figure 15: Harmonic Response 1.5 to 3 Khz

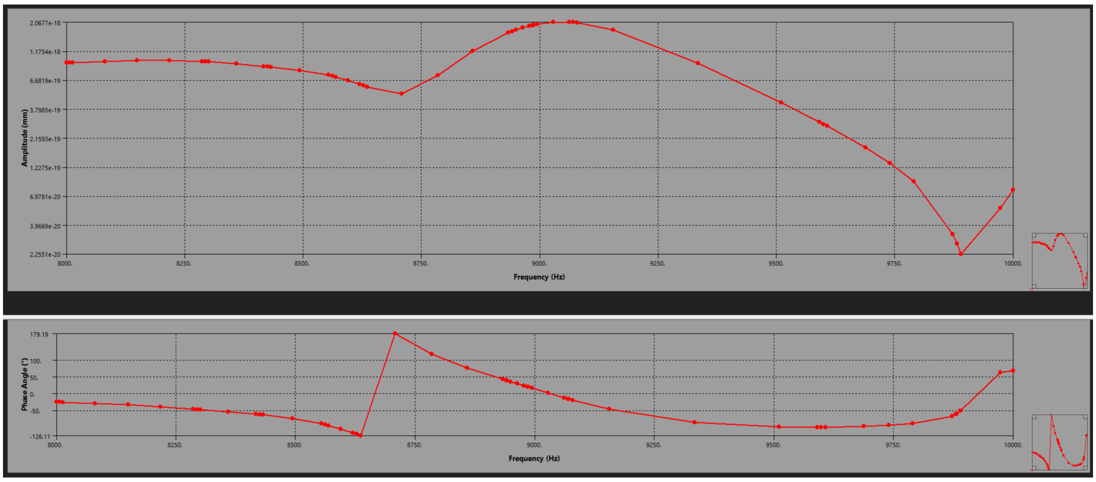


Figure 16: Harmonic Response 8 to 10 Khz

4.6 Summary of Results

Table 7: Summary of analysis results for the brake disc.

Analysis	**Outcome**	**Comments**
Mesh convergence	Passed	$\Delta T_{\max}$ and $\sigma_{\max}$ converged within tolerance; 1 mm global sizing used for all analyses.
Static thermal	Passed	Peak temperature ≈ 257.5°C, well below the 500°C limit for the pad/rotor operating range.
Static structural	Acceptable /Borderline	Peak stress ≈ 348 MPa, close to the 345 MPa yield strength of AISI 420; highest stresses occur locally around cooling holes due to geometric concentrations.
Harmonic response	Passed	No structural resonance observed in the 1–16 kHz band; mode participation smooth with no amplification peaks.

Overall, the disc performs satisfactorily under the defined load case, with the only concern being the elevated local stresses around the cooling holes, which may warrant further geometric refinement.

5 Validation and Discussion

5.1 Researched Validation

Thermal. Both the full-disc model and the 1/10th sector predict a peak steady-state temperature of approximately 257.5 °C at the outer friction track. This value is well below the typical operating limits of stainless-steel motorcycle rotors and regular quality pad compounds, which can tolerate sustained temperatures in the 250–350 °C range [11, 8]. The agreement between the full model and the cyclic-symmetry sector model also validates the thermal boundary conditions and confirms that the reduced model accurately reproduces the thermal response of the full disc.

Structural The strong localisation of stress around cooling-hole edges observed in this study is entirely consistent with published research on ventilated rotors. Park et al. [20] performed a detailed thermo-mechanical analysis of brake discs containing drilled holes and ventilation slits, and reported that "the holes become stress points and potentially cause cracks," with the highest thermal and mechanical stresses occurring precisely at hole boundaries. Their FEA results (Shown in Fig 17 [20]) show distinct stress spikes at the upstream and downstream edges of each hole, closely matching the local maxima identified in the present disc.

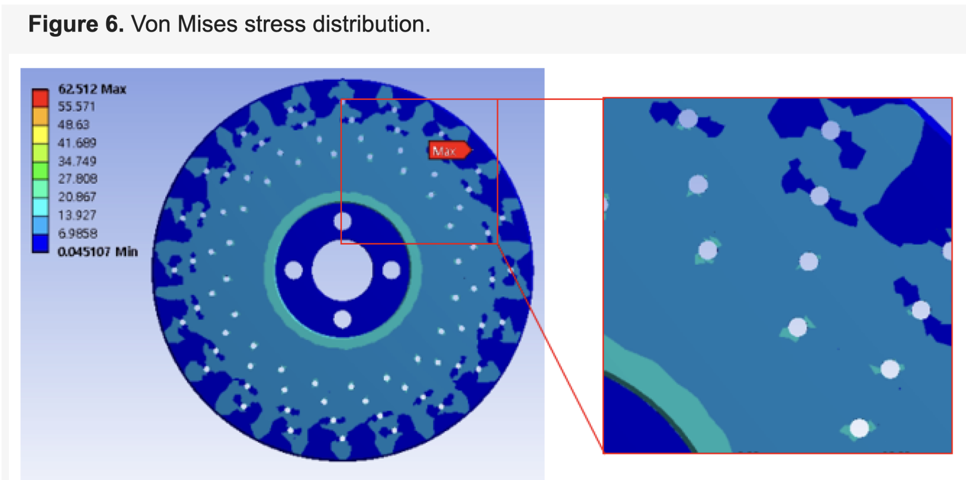


Figure 17: Parks Rotor and Stress Results

In this reports model, although the converged average stress across the friction track is relatively low (∼96 MPa), a local peak of approximately 347 MPa was found at the edge of a cooling hole. Park et al.'s findings provide a clear explanation: these elevated stresses arise not from excessive loading, but from geometric discontinuities that amplify stresses locally even when global loads are moderate. Their optimisation study further demonstrates that modifying hole size, spacing, and edge shape can reduce peak stresses by 15–25%, reinforcing the view that such concentrations are inherent to drilled or ventilated rotor geometries.

Overall, the published literature confirms that the stress behaviour of the present model is physically realistic. The 347 MPa peak is best interpreted as a geometric stress concentration consistent with prior studies, rather than as evidence of structural overload. This comparison validates the thermo-mechanical response predicted by the FEA and highlights geometric refinement—especially hole-edge filleting or alternative cooling patterns—as the most effective direction for reducing local stress magnitudes.

Harmonic Response. According to a chapter on brake squeal from ScienceDirect, brake squeal is commonly classified into low-frequency squeal (1–4 kHz), involving vibration of the rotor, stator, caliper, and axle components, and high-frequency squeal (4–20 kHz), dominated primarily by the rotor–stator pair [21]. The harmonic response of the present disc shows minor amplitude features around 2–3 kHz and 8–10 kHz, which lie directly within these known squeal bands. However, these features do not exhibit the sharp amplitude spikes or 180° phase reversals typically associated with mode-coupling resonance. This indicates that, although the disc responds dynamically within recognised squeal frequency ranges, it does not demonstrate the unstable vibrational behaviour required to generate brake squeal under the simulated operating conditions.

5.2 Calculated Validation

The stress and deformation predictions are the primary focus of the verification, as the stress results were close to the allowable limits for the material. In contrast, the modal frequencies and temperature distributions fall well within acceptable operating ranges and therefore do not require the same level of scrutiny.

5.2.1 Analytical estimate of thermal expansion

As a simple validation of the deformation FEA results, the radial expansion of the disc can be estimated using the 1D thermal strain relation. Thermal strain is defined as the free expansion produced by a temperature change, as summarised by ScienceDirect [17].

$$\varepsilon_T = \alpha \Delta T,$$

where α is the coefficient of thermal expansion and ΔT is the temperature rise relative to the reference temperature used in the structural analysis.

Assuming the outer edge of the disc heats from 25°C to 257.5°C as seen in the thermal results, the temperature increase is

$$\Delta T = 257.5 - 25 \approx 233 \text{ K}.$$

For AISI 420 stainless steel, the coefficient of thermal expansion is $\alpha \approx 10.8 \times 10^{-6}$ K^{-1} (Tab. 4) , and the outer radius of the disc is $r_o = 0.145$ m. Assuming the disc is free to expand, which it is, the radial displacement at the outer edge would be

$$u_r^{\text{free}} = \varepsilon_T r_o = \alpha \Delta T \, r_o = \left(10.8 \times 10^{-6}\right)(233)\,(0.145) = 3.65 \times 10^{-4}\,\text{m} = 0.365 \text{ mm}.$$

The FEA prediction for the radial displacement at the outer edge of the disc is approximately 0.325 mm (see Fig. 11), which compares closely with the analytical estimate of 0.365 mm. This agreement confirms that the thermoelastic expansion of the disc is being captured accurately by the numerical model.

5.2.2 Analytical estimate of thermal stress

A simple upper-bound estimate of the thermal stress can be obtained by considering the outer region of the disc as a ring that attempts to expand relative to the cooler inner region. If the additional expansion associated with a temperature difference ΔT_{eff} is partially restrained, the induced hoop stress can be approximated by

$$\sigma_\theta \approx E\,\alpha\,\Delta T_{\text{eff}},$$

where E is Young's modulus and ν is Poisson's ratio. The thermal stress estimate follows the standard expression for constrained thermal expansion [19].

The full temperature rise $\Delta T = T_{\text{outer}} - T_{\text{ref}}$ is used for estimating free radial expansion, since the disc is largely unconstrained in the radial direction. In contrast, the thermal stress estimate uses an effective temperature difference $\Delta T_{\text{eff}} = T_{\text{outer}} - T_{\text{inner}}$ to represent the differential expansion between the hot outer ring and the cooler inner region, which is what generates the hoop stress.

From the steady-state thermal field, the outer friction track is approximately $T_{\text{outer}} \approx$ 257.5°C, while the inner region of the disc isaround $T_{\text{inner}} \approx$ 123°C (Refer to Fig. 7), giving an effective temperature difference

$$\Delta T_{\text{eff}} \approx T_{\text{outer}} - T_{\text{inner}} \approx 134.5 \text{ K}.$$

Using $E = 200$ GPa and $\alpha = 10.8 \times 10^{-6}$ K^{-1},

$$\sigma_\theta \approx (200 \times 10^9)(10.8 \times 10^{-6})(134.5) = 2.905 \times 10^8 \text{ Pa} = 290.5 \text{ MPa}.$$

This analytical estimate is close to the peak von Mises stress of approximately 347 MPa obtained from the FEA, providing reasonable validation that the magnitude of the thermo-mechanical stresses in the model is consistent with simple thermoelastic theory and suggests that the high stresses around the cooling holes are potentially accurate.

It is important to emphasise that this approach is a deliberate simplification: the actual brake disc is a plate with vents, holes, varying stiffness, and non-uniform temperature fields. Modelling it as a uniform ring neglects these geometric and structural complexities. Nevertheless, this simplified expression still provides useful insight into the order of magnitude of thermal stresses and offers a reasonable validation check against the finite element results.

6 Conclusions

Initial operating conditions were established for a sustained downhill braking scenario, and the disc geometry was modelled from realistic motorcycle rotor dimensions [5]. Finite Element simulations were then conducted to assess the disc's thermal, structural, and vibrational performance against the criteria defined in Table 1.

The steady-state thermal analysis predicted a peak temperature of approximately 257.5 °C at the outer edge of the disc. This value lies within the expected operating range for stainless-steel motorcycle rotors and pad materials, indicating that the rotor's thermal performance under continuous braking is acceptable. The good agreement between the full-disc model and the 1/10th cyclic-symmetry model further validates the accuracy of the applied thermal boundary conditions.

The static structural results showed a maximum von Mises stress of roughly 347 MPa occurring at the edges of the cooling holes. These peak values originate from geometric stress concentrations and may represent local singularities rather than physically meaningful stresses, however, futher investigation is needed to reduce this. Away from these features, the stress levels converge to an average value of approximately 96 MPa, which agrees well with analytical thermoelastic estimates. This suggests that the overall thermo-mechanical behaviour of the disc is reasonable, although the cooling-hole geometry may require refinement to reduce localised stress magnitudes.

The modal and harmonic analyses showed no structural resonances within the 1–16 kHz brake-squeal band. The harmonic response displayed only mild amplitude variations without the sharp peaks or phase reversals characteristic of resonance, indicating low likelihood of rotor-induced squeal for the current design and loading scenario.

Overall, the brake disc performs satisfactorily under the defined thermal and vibrational criteria, with the primary concern being the elevated local stresses around the cooling holes. If analysis were to be continued, future work will focus on refining the cooling-hole geometry, improving boundary conditions, incorporating more realistic caliper loading, and performing transient thermo-mechanical simulations to more accurately capture the rotor's behaviour under more realistic braking conditions.

Acknowledgements

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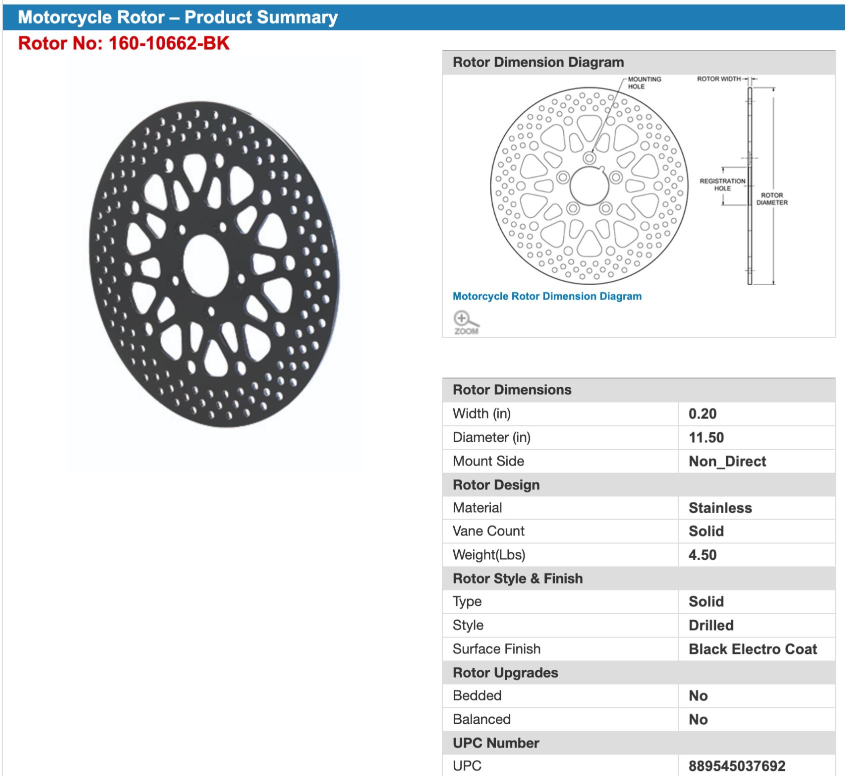
A Appendix

A.1 Source Rotor Specification

The geometric parameters for the simulated ventilated brake disc were adapted from the Disc Brake (Part No. 160-10662-BK) manufactured by Wilwood Engineering. Key dimensional and design characteristics were referenced from the official product page.



Rotor Dimensions	
Width (in)	0.20
Diameter (in)	11.50
Mount Side	Non_Direct
Rotor Design	
Material	Stainless
Vane Count	Solid
Weight(Lbs)	4.50
Rotor Style & Finish	
Type	Solid
Style	Drilled
Surface Finish	Black Electro Coat
Rotor Upgrades	
Bedded	No
Balanced	No
UPC Number	
UPC	889545037692

Figure 18: Wilwood (Part No. 160-10662-BK) [3].

A.2 Mesh Convergence Table

Table 8: Updated mesh refinement results showing variation of stress, temperature, and element count with changing mesh size.

Element Size (mm)	No. Elements (–)	Max Stress (MPa)	Avg Stress (MPa)	Max Temp (°C)
5	266	233.50	99.62	248.56
4	358	210.91	100.35	250.83
3	577	239.48	100.97	253.90
2	1140	293.45	100.22	255.84
1	4605	345.59	97.90	257.50
0.5	18372	356.69	96.71	257.93
0.25	73454	361.62	96.23	258.04